run perpendicular to streets, that are also 100 meters long. Just before class, Marge and Lisa compare their designs and realize that something is terribly wrong. After appropriately configuring her grid so that the Monument lies in the center, Marge finds that Kwik-E-Mart is at 8th Street and 5th Avenue and the nuclear power plant is at 3rd Street and 5th Avenue, as shown in Figure 3.2a. But in Lisa's design, the addresses are completely different: the Kwik-E-Mart is near the corner of 7th Street and 3rd Avenue, while the power plant is at 4th Street and 7th Avenue, as in Figure 3.2b. Clearly, someone has made a mistake.

After a moment's thought, though, Lisa realizes what's going on. There are no mistakes. She and Marge are both right. They merely chose different orientations for their street and avenue grids. Marge's streets and avenues run at an angle relative to Lisa's; their grids are rotated relative to each other; they have sliced up Springfield into streets and avenues in two different ways (see Figure 3.2c). The lesson here is simple, yet important. There is freedom in how Springfield—a region of space—can be organized by streets and avenues. There are no "absolute" streets or "absolute" avenues. Marge's choice is as valid as Lisa's—or any other possible orientation, for that matter.

Hold this idea in mind as we paint time into the picture. We are used to thinking about space as the arena of the universe, but physical processes occur in some region of space *during some interval of time.* As an example, imagine that Itchy and Scratchy are having a duel, as illustrated in Figure 3.3a, and the events are recorded moment by moment in the fashion of one of those old-time flip books. Each page is a "time slice"—like a still frame in a filmstrip—that shows what happened in a region of space at one moment of time. To see what happened at a different moment of time you flip to a different page.[4] (Of course, space is three-dimensional while the pages are two-dimensional, but let's make this simplification for ease of thinking and drawing figures. It won't compromise any of our conclusions.) By way of terminology, a region of space considered over an interval of time is called a region of *spacetime;* you can think of a region of spacetime as a record of all things that happen in some region of space during a particular span of time.